

Student's name

Student's number

Teacher's name



PLC PRESBYTERIAN
LADIES' COLLEGE
SYDNEY
1888

2013
TRIAL
HIGHER SCHOOL CERTIFICATE
EXAMINATION

Mathematics

General Instructions

- Reading time - 5 minutes
- Working time - 3 hours
- Write using blue or black pen
Black is preferred
- Board-approved calculators may be used
- A table of standard integrals is provided at the back of this paper
- All necessary working should be shown in every question

Total Marks – 100

Section I: Pages 3-5

10 marks

- Attempt questions 1-10, using the answer sheet on page 15.
- Allow about 15 minutes for this section

Section II: Pages 6-11

90 marks

- Attempt questions 11-16, using the booklets provided.
- Allow about 2 hours 45 minutes for this section

Multiple Choice	11	12	13	14	15	16	Total
							%

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Section I

10 marks

Attempt Questions 1–10

Allow about 15 minutes for this section

1. What is $\frac{4}{e}$ correct to 3 significant figures?

(A) 1.47
(B) 1.470
(C) 1.471
(D) 1.472

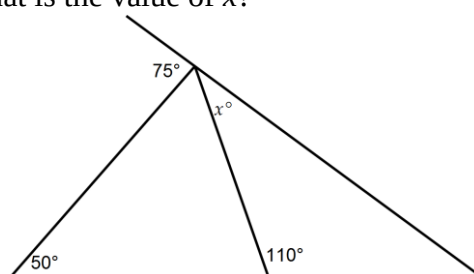
2. What is the gradient of any line perpendicular to the line $3x - 2y + 12 = 0$?

(A) $-\frac{3}{2}$
(B) $-\frac{2}{3}$
(C) $\frac{2}{3}$
(D) $\frac{3}{2}$

3. What is the value of y if $\frac{1}{3 - \sqrt{2}} = x + y\sqrt{2}$?

(A) -1
(B) $-\frac{1}{7}$
(C) $\frac{1}{7}$
(D) 1

4. In the diagram, what is the value of x ?

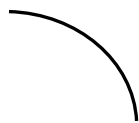


(A) 45
(B) 50
(C) 60
(D) 65

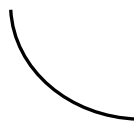
5. Which diagram below represents the following statements?

$$f'(x) < 0 \text{ and } f''(x) > 0$$

(A)



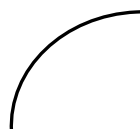
(B)



(C)



(D)



6. Solve $\sin 2x = -\frac{1}{2}$ for $0 \leq x \leq 2\pi$.

(A) $x = \frac{\pi}{12}, \frac{5\pi}{12}$

(B) $x = \frac{7\pi}{12}, \frac{11\pi}{12}$

(C) $x = \frac{\pi}{12}, \frac{5\pi}{12}, \frac{13\pi}{12}, \frac{17\pi}{12}$

(D) $x = \frac{7\pi}{12}, \frac{11\pi}{12}, \frac{19\pi}{12}, \frac{23\pi}{12}$

7. The number of ants in my Ant Farm after t days is given by $A = A_0 e^{kt}$, where k is a positive constant. Initially there are 50 ants and after 4 days the number has increased to 90. How many days, correct to 3 decimal places, will it take for the number of ants to reach 180?

(A) 0.147

(B) 0.545

(C) 8.000

(D) 8.717

8. A circular pizza with diameter of 28 centimetres is to be cut up into 12 equal sectors. What is the area of one sector, in square centimetres?

- (A) $\frac{49}{3}(\pi - 3)$
(B) $\frac{49\pi}{3}$
(C) 294
(D) 2891

9. For the curve $y = x^3 - x^2 - x + 1$
Which of the following statements are true?

I: Stationary points occur at $x = \frac{-1}{3}$ and $x = 1$

II: The curve is concave down for $x < \frac{1}{3}$

- (A) **I** only
(B) **II** only
(C) Both **I** and **II**
(D) Neither **I** nor **II**

10. Consider the function $y = (4 - x^2)^{\frac{1}{2}}$.

Use the **trapezoidal rule** with 3 function values to find the approximate area under the curve $y = (4 - x^2)^{\frac{1}{2}}$ for $0 \leq x \leq 2$, correct to 2 decimal places. What is this approximate area, in square units?

- (A) 1.24
(B) 1.87
(C) 2.73
(D) 3.14

End of Section I

Section II

90 marks

Attempt Questions 11–16

Allow about 2 hours 45 minutes for this section

Question 11:

15 Marks

- a) Differentiate with respect to x :
- (i) $\frac{\log_e x}{x}$ 2
- (ii) $x \sin(3x-1)$ 2
- b) Evaluate $\int_1^{e^3} 4x^{-1} dx$ 2
- c) ABC is a triangle, right-angled at A . AD is perpendicular to BC . P is a point on the interval BD such that $CP=CA$. If $\angle APC = 80^\circ$, show that PA bisects $\angle BAD$, with reasons. 3
- d) Prove the following
 $(\sec \theta + \tan \theta)^2 = \frac{1 + \sin \theta}{1 - \sin \theta}$ 2
- e) Solve for x : $|2x-5| = 4x-1$ 2
- f) The vertex of a parabola $y = x^2 + bx + c$ is $(-4, -18)$. Find the values of b and c . 2

End of Question 11

Question 12:**15 Marks**

- a)**
- (i) Sketch the curve $y = \sqrt{3} \tan x$ for $-\pi \leq x \leq \pi$. **1**
 - (ii) Hence, or otherwise, solve $\sqrt{3} \tan x > 1$ in this domain. **2**
- b)** For the parabola $x^2 = 16y$
- (i) Find the co-ordinates of the focus, S . **1**
 - (ii) Show that the equation of the tangent to this parabola at the point $P(4,1)$ is $x - 2y - 2 = 0$. **2**
 - (iii) The straight line joining P to the focus, S , intersects the parabola again at R . Show that the co-ordinates of R are $(-16,16)$. **3**
 - (iv) The equation of the tangent at R is $2x + y + 16 = 0$. Show that the tangents at P and R intersect at T on the directrix. **2**
 - (v) Find the distance PT . **1**
 - (vi) Find the perpendicular distance from R to PT . **2**
 - (vii) Hence find the exact area of $\triangle TRP$. **1**

End of Question 12

Question 13:**15 Marks**

- a)** Sketch on a Cartesian plane the region where $x^2 + y^2 \geq 9$ and $y \geq x - 3$ hold simultaneously. **2**
- b)** Give the exact solution of $2e^{-2x} - 5e^{-x} + 2 = 0$. **2**
- c)** (i) Sketch the line $2x - 3y = 6$, showing x and y intercepts. **1**
- (ii) Find the co-ordinates of the point on the straight line $2x - 3y = 6$ which is closest to the origin. **3**
- d)** For the quadratic $kx^2 - (2k + 1)x + 2 = 0$, find k if:
- (i) $x = 1$ is a root of the quadratic **1**
- (ii) Roots are real **3**
- (iii) Roots are equal in magnitude but opposite in sign **2**
- (iv) One root is the reciprocal of the other **1**

End of Question 13

Question 14:

15 Marks

- a) Find $\int e^{5x+2} dx$ 1
- b) The curve $y = x^3$ is rotated around the ***y*-axis** from $y = -1$ to $y = 1$. Find the volume of the solid of revolution. 3
- c) Sketch $y = 1 - \cos \frac{2x}{3}$ for $0 \leq x \leq 3\pi$. 3
- d) If $y = \frac{x}{\sin x}$, show that $\sin x \frac{dy}{dx} + y \cos x = 1$. 2
- e) Two circles, with centres at A and B , intersect at C and D as shown in the diagram below. The radius of the circle centred at A is 2 metres, $CA \perp CB$ and $AB = 4$ metres.

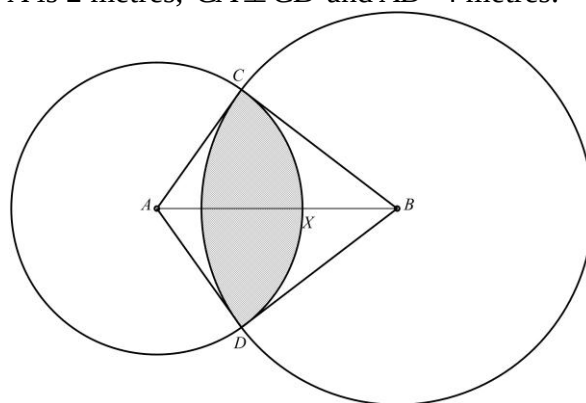


Diagram NOT
drawn to scale

- (i) Copy or trace the diagram into your answer booklet.
- (ii) Prove $\triangle ABC \equiv \triangle ABD$. 2
- (iii) Given the arc length CXD is $\frac{2\pi}{3}$ metres, show that $\angle CAB = \frac{\pi}{6}$. 1
- (iv) Find the shaded area. 3

End of Question 14

Question 15:**15 Marks**

- a)** A variable line with gradient m passes through the point $(1, 2)$ and intersects the x -axis at $A(a, 0)$ and the y -axis at $B(0, b)$, where both a and b are positive.
- (i) Show that the equation of AB is $y = mx + 2 - m$ **1**
- (ii) Show that the area of triangle $\triangle AOB$ is given by **2**
$$\text{Area} = \frac{4m - m^2 - 4}{2m}$$
- (iii) Find the smallest possible area of $\triangle AOB$. **2**
- b)** The displacement, x metres, of a particle moving along the x -axis is given at time t seconds by the equation $x = 12e^{-2t} - 12 + 24t$.
- (i) Show the particle is initially at rest. **1**
- (ii) Find the velocity when $t = \log_e \left(\frac{4}{3} \right)$ seconds. **2**
- (iii) What is the limit approached by the velocity as t increases? **1**
- (iv) Sketch the velocity-time graph for $0 \leq t \leq 3$. On your sketch clearly show the velocity at $t=1.5$. **2**
- (v) Using Simpson's Rule with 3 function values, find the distance travelled in the first 3 seconds. Write your answer correct to 1 decimal place. **2**
- (vi) Prove that the acceleration is $48-2v$, where v is velocity. **2**

End of Question 15

Question 16**15 Marks**

- a) Find the values of A and B if $\int_{\frac{\pi}{4}}^{\frac{\pi}{2}} \cot x \, dx = A \log_e B$ 2
- b) For the curve $y = f(x)$ where $f(x) = \frac{\sec^2 x}{1 + \tan x}$ for $0 \leq x \leq \pi$,
- (i) Find the y -intercept and $f(\pi)$. 2
- (ii) Show that $\frac{dy}{dx} = \frac{\sec^2 x (\tan^2 x + 2 \tan x - 1)}{(1 + \tan x)^2}$ 3
- (iii) There are two stationary points on the curve $y = \frac{\sec^2 x}{1 + \tan x}$ in the interval $0 \leq x \leq \pi$. Given $(0.39, 0.83)$ is a minimum stationary point, find the other stationary point and determine its nature. 3
- Do not try to find the points of inflexion.*
- (iv) Show there are vertical asymptotes at $x = \frac{\pi}{2}$ and $x = \frac{3\pi}{4}$ 1
- (v) Sketch the curve in the interval $0 \leq x \leq \pi$, clearly showing all the above information. 2
- (vi) Find the exact area under the curve $f(x) = \frac{\sec^2 x}{1 + \tan x}$ bounded by the x -axis, the y -axis and $x = \frac{\pi}{3}$ in exact form. 2

End of Paper

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STANDARD INTEGRALS

$$\int x^n dx = \frac{1}{n+1} x^{n+1}, \quad n \neq -1; \quad x \neq 0, \text{ if } n < 0$$

$$\int \frac{1}{x} dx = \ln x, \quad x > 0$$

$$\int e^{ax} dx = \frac{1}{a} e^{ax}, \quad a \neq 0$$

$$\int \cos ax dx = \frac{1}{a} \sin ax, \quad a \neq 0$$

$$\int \sin ax dx = -\frac{1}{a} \cos ax, \quad a \neq 0$$

$$\int \sec^2 ax dx = \frac{1}{a} \tan ax, \quad a \neq 0$$

$$\int \sec ax \tan ax dx = \frac{1}{a} \sec ax, \quad a \neq 0$$

$$\int \frac{1}{a^2 + x^2} dx = \frac{1}{a} \tan^{-1} \frac{x}{a}, \quad a \neq 0$$

$$\int \frac{1}{\sqrt{a^2 - x^2}} dx = \sin^{-1} \frac{x}{a}, \quad a > 0, \quad -a < x < a$$

$$\int \frac{1}{\sqrt{x^2 - a^2}} dx = \ln \left(x + \sqrt{x^2 - a^2} \right), \quad x > a > 0$$

$$\int \frac{1}{\sqrt{x^2 + a^2}} dx = \ln \left(x + \sqrt{x^2 + a^2} \right)$$

NOTE : $\ln x = \log_e x$, $x > 0$

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Mathematics: Multiple Choice Answer Sheet

Student Number_____

Completely fill the response oval representing the most correct answer.

- | | | | | | | | | |
|-----|---|-----------------------|---|-----------------------|---|-----------------------|---|-----------------------|
| 1. | A | ☐ | B | ☐ | C | ☐ | D | ☐ |
| 2. | A | ☐ | B | ☐ | C | ☐ | D | ☐ |
| 3. | A | ☐ | B | ☐ | C | ☐ | D | ☐ |
| 4. | A | ☐ | B | ☐ | C | ☐ | D | ☐ |
| 5. | A | ☐ | B | ☐ | C | ☐ | D | ☐ |
| 6. | A | ☐ | B | ☐ | C | ☐ | D | ☐ |
| 7. | A | ☐ | B | ☐ | C | ☐ | D | ☐ |
| 8. | A | ☐ | B | ☐ | C | ☐ | D | ☐ |
| 9. | A | ☐ | B | ☐ | C | ☐ | D | ☐ |
| 10. | A | ☐ | B | ☐ | C | ☐ | D | ☐ |

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Mathematics: Multiple Choice Answer Sheet

Student Number ANSWERS

Completely fill the response oval representing the most correct answer.

- | | | | | | | | | |
|-----|---|----------------------------------|---|----------------------------------|---|----------------------------------|---|----------------------------------|
| 1. | A | ☒ | B | ☐ | C | ☐ | D | ☐ |
| 2. | A | ☐ | B | ☒ | C | ☐ | D | ☐ |
| 3. | A | ☐ | B | ☐ | C | ☒ | D | ☐ |
| 4. | A | ☒ | B | ☐ | C | ☐ | D | ☐ |
| 5. | A | ☐ | B | ☒ | C | ☐ | D | ☐ |
| 6. | A | ☐ | B | ☐ | C | ☐ | D | ☒ |
| 7. | A | ☐ | B | ☐ | C | ☐ | D | ☒ |
| 8. | A | ☐ | B | ☒ | C | ☐ | D | ☐ |
| 9. | A | ☐ | B | ☐ | C | ☒ | D | ☐ |
| 10. | A | ☐ | B | ☐ | C | ☒ | D | ☐ |

Academic Year		Calendar Year	
Course		Name of task/exam	

Question 11:

$$a \quad i \quad \frac{d}{dx} \left(\frac{\log_e x}{x} \right)$$

$$= \frac{v u' - u v'}{v^2}$$

$$= \frac{x \left(\frac{1}{x} \right) - \ln x (1)}{x^2}$$

$$= \frac{1 - \ln x}{x^2}$$

quotient rule

$$u = \ln x$$

$$u' = \frac{1}{x}$$

$$v = x$$

$$v' = 1$$

$$ii \quad \frac{d}{dx} [x \sin(3x-1)]$$

product rule

$$u = x$$

$$u' = 1$$

$$v = \sin(3x-1)$$

$$v' = 3 \cos(3x-1)$$

$$= x(3 \cos(3x-1)) + \sin(3x-1)(1)$$

$$= 3x \cos(3x-1) + \sin(3x-1)$$

$$b \quad \int_1^{e^3} 4x^{-1} dx$$

$$= 4 \int_1^{e^3} \frac{1}{x} dx$$

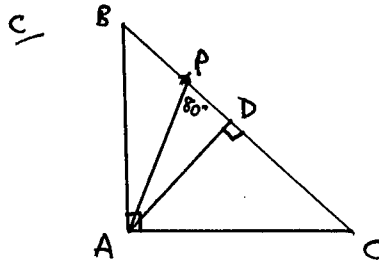
$$= 4 [\ln x]_1^{e^3}$$

$$= 4 [\ln e^3 - \ln 1]$$

$$= 4 [3 \ln e - 0]$$

$$= 4(3)$$

$$= 12$$



In $\triangle PCA$: $PC = CA$ (given)

$\angle APC = \angle PAC$ (angles opposite equal sides are equal).
 $80^\circ = \angle PAC$

$$\therefore \angle PAC = 80^\circ$$

In $\triangle ABC$: $\angle BAC = 90^\circ$ (given)

$$\therefore \angle BAP = \angle BAC - \angle PAC \quad (\text{right angle})$$

$$= 90^\circ - 80^\circ$$

$$= 10^\circ$$

In $\triangle PDA$: $\angle PDA = 90^\circ$ ($AD \perp BC$)

$$\therefore \angle PAD = 180 - \angle DPA - \angle PDA \quad (\text{angle sum of triangle})$$

$$= 180^\circ - 80^\circ - 90^\circ$$

$$= 10^\circ$$

$$\therefore \angle BAP = \angle PAD$$

$\therefore PA$ bisects $\angle BAD$.

d/ Prove

$$(\sec \theta + \tan \theta)^2 = \frac{1 + \sin \theta}{1 - \sin \theta}$$

$$\text{LHS} = (\sec \theta + \tan \theta)^2$$

$$= \sec^2 \theta + 2 \sec \theta \tan \theta + \tan^2 \theta$$

Academic Year		Calendar Year	
Course		Name of task/exam	

$$\text{LHS} = \frac{1}{\cos^2 \theta} + \frac{2 \times \frac{1}{\cos \theta} \times \frac{\sin \theta}{\cos \theta}}{\cos^2 \theta} + \frac{\sin^2 \theta}{\cos^2 \theta}$$

$$= \frac{1 + 2\sin \theta + \sin^2 \theta}{\cos^2 \theta}$$

$$= \frac{(1 + \sin \theta)^2}{(1 - \sin^2 \theta)}$$

$$= \frac{(1 + \sin \theta)(\cancel{1 + \sin \theta})}{(1 - \sin \theta)(\cancel{1 + \sin \theta})}$$

$$= \frac{1 + \sin \theta}{1 - \sin \theta}$$

$$= \text{RHS}$$

$\therefore$ proved

$$\underline{e} \quad |2x - 5| = 4x - 1$$

$$2x - 5 = 4x - 1$$

$$-2x = 4$$

$$x = -2$$

$$\begin{aligned} \underline{\text{Check}} \quad \text{LHS} &= |2(-2) - 5| \\ &= |-9| \\ &= 9 \end{aligned}$$

$$\begin{aligned} \text{RHS} &= 4(-2) - 1 \\ &= -9 \end{aligned}$$

$$\text{LHS} \neq \text{RHS}$$

$\therefore$ not a solution

$$\underline{\text{OR}} \quad -(2x - 5) = 4x - 1$$

$$-2x + 5 = 4x - 1$$

$$-6x = -6$$

$$x = 1$$

$$\text{LHS} = |2(1) - 5|$$

$$= |-3|$$

$$= 3$$

$$\text{RHS} = 4(1) - 1$$

$$= 3$$

$$\text{LHS} = \text{RHS}$$

$\therefore x = 1$ is solution.

$$\underline{f} \quad y = x^2 + bx + c$$

$$\text{Vertex: } x = \frac{-b}{2(1)}$$

$$-4 = \frac{-b}{2}$$

$$-8 = -b$$

$$\therefore b = 8$$

$(-4, -18)$ satisfies eqn. of parabola

$$-18 = (-4)^2 + 8(-4) + c$$

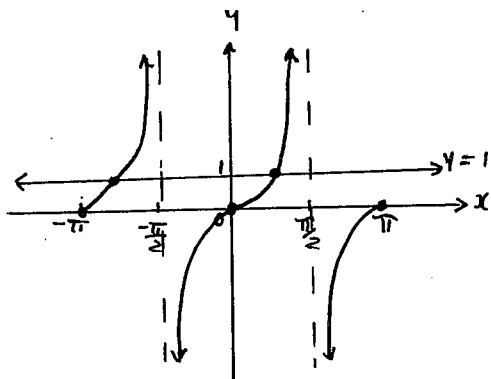
$$-18 = 16 - 32 + c$$

$$-18 = -16 + c$$

$$c = -2$$

$$\therefore b = 8 \quad c = -2$$

Academic Year		Calendar Year	
Course		Name of task/exam	

Question 12 :

ii solve $\sqrt{3} \tan x = 1$
 $\tan x = \frac{1}{\sqrt{3}}$ ✓
 $x = \frac{\pi}{6}, -\frac{5\pi}{6}$

$\therefore -\frac{5\pi}{6} < x < -\frac{\pi}{2}, \frac{\pi}{6} < x < \frac{\pi}{2}$

b $x^2 = 16y$ vertex $(0, 0)$

i $4a = 16$
 $a = 4$

$\therefore S(0, 4)$

ii $y = \frac{x^2}{16}$

$\frac{dy}{dx} = \frac{2x}{16}$

$\frac{dy}{dx} = \frac{x}{8}$

at $P(4, 1)$

$m_{\text{tangent}} = \frac{4}{8}$

$m = \frac{1}{2}$

$\therefore$ eqn tangent:

$y - y_1 = m(x - x_1)$

$y - 1 = \frac{1}{2}(x - 4)$

$2y - 2 = x - 4$

$0 = x - 2y - 4 + 2$

$\therefore x - 2y - 2 = 0$

iii equation of line joining P and S:

$m = \frac{4-1}{0-4}$

$= \frac{3}{-4}$

eqn: $y - 4 = -\frac{3}{4}(x - 0)$

$4y - 16 = -3x$

$3x + 4y - 16 = 0$

To find R: solve simult eqns

$x^2 = 16y$ ①

$3x + 4y - 16 = 0$ ②

① $y = \frac{x^2}{16}$

$3x + 4\left(\frac{x^2}{16}\right) - 16 = 0$

$3x + \frac{x^2}{4} = 16$

$x^2 + 12x - 64 = 0$

$(x - 4)(x + 16) = 0$

$x = 4 \quad x = -16$

$\therefore (4, 1) \quad (-16, 16)$

$\therefore R(-16, 16)$

Academic Year		Calendar Year	
Course		Name of task/exam	

iv egn tangent R

$$2x + y + 16 = 0$$

egn tangent P

$$x - 2y - 2 = 0$$

Solve to find y:

(egn directrix $y = -4$).

$$2x + y + 16 = 0$$

$$2x - 4y - 4 = 0$$

$$5y + 20 = 0$$

$$y = -4.$$

$\therefore$ T lies on directrix

$\therefore$ coordinates of T $(-6, -4)$

$$d_{PT} = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

$$= \sqrt{(4 - (-6))^2 + (1 - (-4))^2}$$

$$= \sqrt{100 + 25}$$

$$= \sqrt{125}$$

$$= 5\sqrt{5} \text{ units}$$

vi R $(-16, 16)$

egn PT is egn of tangent at P

$$x - 2y - 2 = 0$$

$$\therefore d_{\perp} = \left| \frac{ax_1 + by_1 + c}{\sqrt{a^2 + b^2}} \right|$$

$$= \left| \frac{1(-16) - 2(16) - 2}{\sqrt{1^2 + (-2)^2}} \right|$$

$$d_{\perp} = \left| \frac{-16 - 32 - 2}{\sqrt{5}} \right|$$

$$= \frac{50}{\sqrt{5}}$$

$$= 10\sqrt{5} \text{ units}$$

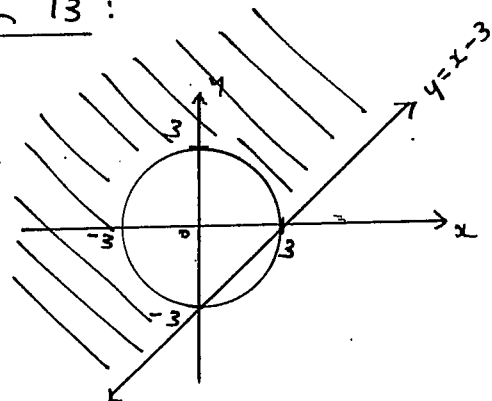
$$\text{vii } A = \frac{1}{2} b h$$

$$= \frac{1}{2} (5\sqrt{5}) (10\sqrt{5})$$

$$= 125 \text{ units}^2$$

Question 13:

a



$$x^2 + y^2 \geq 9$$

outside circle

$$y \geq x - 3$$

Test $(3,3)$
 $3 \geq 0$
true

$$b \quad 2e^{-2x} - 5e^{-x} + 2 = 0$$

$$\text{Let } u = e^{-x}$$

$$2u^2 - 5u + 2 = 0$$

$$(2u - 1)(u - 2) = 0$$

$$u = \frac{1}{2} \quad u = 2$$

$$e^{-x} = \frac{1}{2}$$

$$e^{-x} = 2$$

$$-x = \ln\left(\frac{1}{2}\right)$$

$$-x = \ln 2$$

$$x = -\ln\left(\frac{1}{2}\right)$$

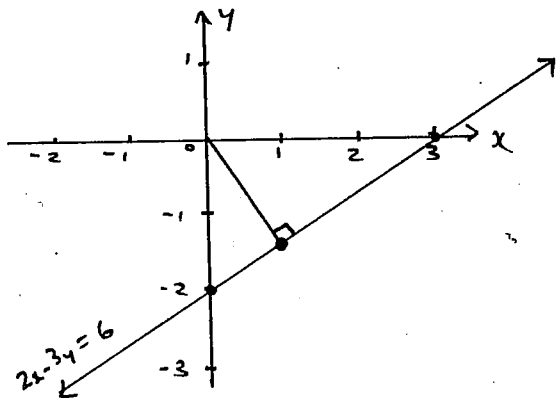
$$x = -\ln 2$$

$$\text{or } x = \ln 2.$$

$$x = \ln\left(\frac{1}{2}\right)$$

Academic Year		Calendar Year	
Course		Name of task/exam	

c. i.



$$2x - 3y = 6$$

x-intercept: set $y = 0$

$$2x = 6$$

$$x = 3$$

y-intercept: set $x = 0$

$$-3y = 6$$

$$y = -2$$

ii. Closest to origin.

We need to find eqn of line passing through origin and perpendicular to $2x - 3y = 6$.

$$\therefore m_{\text{line}} = \frac{2}{3}$$

$$m_{\perp} = -\frac{3}{2} \quad (0, 0)$$

$$y - 0 = -\frac{3}{2}(x - 0)$$

$$2y = -3x$$

Need to solve $2x - 3y = 6$ ①

$$3x + 2y = 0$$
 ②

$$6x - 9y = 18$$
 ① $\times 3$

$$6x + 4y = 0$$
 ② $\times 2$

$$-13y = 18$$
 ① - ②

$$y = -\frac{18}{13}$$

$$x = \frac{12}{13}$$

$$\therefore \text{pt is } \left(\frac{12}{13}, -\frac{18}{13} \right)$$

$$d. \quad kx^2 - (2k+1)x + 2 = 0$$

i. $x = 1$ is root

$\therefore$ it satisfies quadratic

$$k(1)^2 - (2k+1)(1) + 2 = 0$$

$$k - 2k - 1 + 2 = 0$$

$$-k + 1 = 0$$

$$k = 1$$

ii. for real roots $\Delta \geq 0$

$$\therefore b^2 - 4ac \geq 0$$

$$[-(2k+1)]^2 - 4(k)(2) \geq 0$$

$$4k^2 + 4k + 1 - 8k \geq 0$$

$$4k^2 - 4k + 1 \geq 0$$

$$(2k - 1)^2 \geq 0$$

for real roots all k satisfy.

iii. Let roots be $\alpha, -\alpha$

Sum of roots:

$$\alpha + -\alpha = -\frac{b}{a}$$

$$0 = \frac{2k+1}{k}$$

$$2k+1 = 0$$

$$k = -\frac{1}{2}$$

iv. let roots be $\alpha, \frac{1}{\alpha}$

Product of roots:

$$\alpha \left(\frac{1}{\alpha} \right) = \frac{c}{a}$$

$$1 = \frac{2}{k}$$

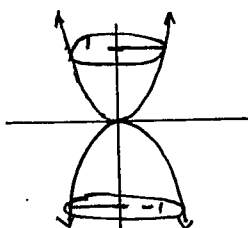
$$\therefore k = 2$$

Academic Year		Calendar Year	
Course		Name of task/exam	

Question 14 :

$$a \quad \int e^{5x+2} dx = \frac{e^{5x+2}}{5} + C$$

$$b \quad y = x^3$$



$$V = \pi \times 2 \int_0^1 x^2 dy$$

$$x = y^{\frac{1}{3}}$$

$$\therefore V = 2\pi \int_0^1 \left(y^{\frac{1}{3}}\right)^2 dy$$

$$= 2\pi \left[\frac{3y^{\frac{5}{3}}}{5} \right]_0^1$$

$$= 2\pi \left[\frac{3}{5} (1)^{\frac{5}{3}} - \frac{3}{5} (0)^{\frac{5}{3}} \right]$$

$$= 2\pi \times \frac{3}{5}$$

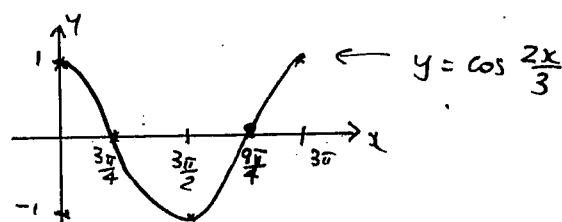
$$= \frac{6\pi}{5} \text{ units}^3$$

$$\leq y = 1 - \cos \frac{2x}{3} \quad 0 \leq x \leq 3\pi$$

$$\text{Consider } y = \cos \frac{2x}{3}$$

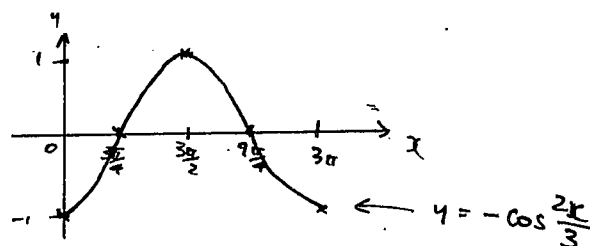
$$\text{period} = \frac{2\pi}{2/3} = 3\pi$$

$$\text{amplitude} = 1.$$



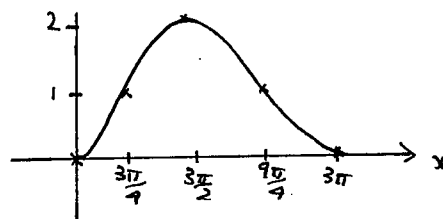
$$\text{Consider } y = -\cos \frac{2x}{3}$$

reflection in x-axis



$$\text{Consider } y = 1 - \cos \frac{2x}{3}$$

shifted up 1 unit



Academic Year		Calendar Year	
Course		Name of task/exam	

d $y = \frac{x}{\sin x}$

$$\frac{dy}{dx} = \frac{vu' - uv'}{v^2}$$

$$\begin{aligned} u &= x \\ u' &= 1 \\ v &= \sin x \\ v' &= \cos x \end{aligned}$$

$$\frac{dy}{dx} = \frac{\sin x (1) - x (\cos x)}{(\sin x)^2}$$

$$= \frac{\sin x - x \cos x}{\sin^2 x}$$

RTS $\sin x \frac{dy}{dx} + y \cos x = 1$

$$\text{LHS} = \sin x \left[\frac{\sin x - x \cos x}{\sin^2 x} \right] + \frac{x}{\sin x} (\cos x)$$

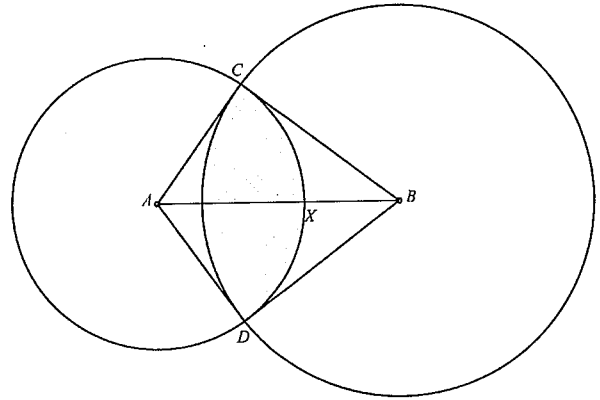
$$= \frac{\sin x - x \cos x}{\sin x} + \frac{x \cos x}{\sin x}$$

$$= \frac{\sin x}{\sin x}$$

$$= 1$$

$$= \text{RHS}$$

e i



ii

In $\triangle ABC$ and $\triangle ABD$

$AC = AD$ (equal radii)

$BC = BD$ (equal radii)

AB common

$\therefore \triangle ABC \equiv \triangle ABD$ (SSS)

iii $\angle = r \theta$

$$\frac{2\pi}{3} = 2\theta$$

$$\theta = \frac{\pi}{3}$$

$$\therefore \angle CAD = \frac{\pi}{3}$$

$$\therefore \angle CAB = \frac{1}{2} \angle CAD$$

[Since $\angle CAB = \angle DAB$
corresponding angles in congruent
triangles are equal].

$$\therefore \angle CAB = \frac{\pi}{6}$$

Academic Year		Calendar Year	
Course		Name of task/exam	

iv In $\triangle ACB$

$$CA = 2m, AB = 4m$$

$$CA \perp CB$$

$\therefore$ right angled triangle.

$$\therefore 4^2 = 2^2 + (CB)^2$$

$$16 = 4 + (CB)^2$$

$$CB = \sqrt{12}$$

$$CB = 2\sqrt{3}$$

Using circle centre A

$$A_{\text{segment}} = \frac{1}{2} 2^2 \left(\frac{\pi}{3} - \sin \frac{\pi}{3} \right)$$

$$= 2 \left(\frac{\pi}{3} - \frac{\sqrt{3}}{2} \right)$$

Using circle centre B

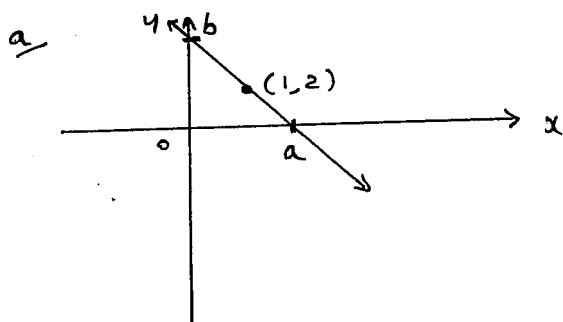
$$A_{\text{segment}} = \frac{1}{2} (2\sqrt{3})^2 \left(\frac{2\pi}{3} - \sin \frac{2\pi}{3} \right)$$

$$= \frac{1}{2} (12) \left(\frac{2\pi}{3} - \frac{\sqrt{3}}{2} \right)$$

$$\therefore A_{\text{shaded}} = \left(\frac{2\pi}{3} - \sqrt{3} \right) + 6 \left(\frac{2\pi}{3} - \frac{\sqrt{3}}{2} \right)$$

$$= \frac{14\pi}{3} - 4\sqrt{3} \quad \text{units}^2$$

Question 15:



$\therefore$ eqn of AB

$$y = mx + b$$

we know (1, 2) lies on line with gradient m

$$2 = m + b$$

$$\therefore b = 2 - m$$

$\therefore$ eqn of AB is

$$y = mx + 2 - m$$

$$\text{ii } A = \frac{1}{2} bh$$

$$= \frac{1}{2} \times a \times b$$

for the line $y = mx + 2 - m$
(a, 0) satisfies

$$0 = ma + 2 - m$$

$$m - 2 = ma$$

$$a = \frac{m - 2}{m}$$

for the line $y = mx + 2 - m$
(0, b) satisfies

$$b = 2 - m$$

$$\therefore A = \frac{1}{2} \left(\frac{m - 2}{m} \right) (2 - m)$$

$$= \frac{(m - 2)(2 - m)}{2m}$$

$$= \frac{2m - m^2 - 4 + 2m}{2m}$$

$$= \frac{4m - m^2 - 4}{2m}$$

Academic Year		Calendar Year	
Course		Name of task/exam	

$$\begin{aligned} \text{iii} \quad \frac{dA}{dm} &= \frac{vu' - uv'}{v^2} & u &= 4m - m^2 - 4 \\ & & u' &= 4 - 2m \\ & & v &= 2m \\ & & v' &= 2 \\ &= \frac{2m(4-2m) - (4m-m^2-4)(2)}{(2m)^2} \\ &= \frac{8m - 4m^2 - 8m + 2m^2 + 8}{(2m)^2} \\ &= \frac{-2m^2 + 8}{(2m)^2} \end{aligned}$$

for max/min $\frac{dA}{dm} = 0$

$$\therefore -2m^2 + 8 = 0$$

$$2m^2 = 8$$

$$m^2 = 4$$

$$m = \pm 2$$

Since a and b both positive
line AB is decreasing $\therefore m = -2$.

$$\begin{array}{c|c|c|c} m & -3 & -2 & -1 \\ \hline \frac{dA}{dm} & -\frac{10}{36} & 0 & \frac{6}{4} \end{array}$$

∴ min.

$$\begin{aligned} \therefore \text{min Area} &= \frac{4(-2) - (-2)^2 - 4}{2(-2)} \\ &= \frac{-8 - 4 - 4}{-4} \\ &= 4 \text{ units}^2 \end{aligned}$$

$$b \quad x = 12e^{-2t} - 12 + 24t$$

i initially $t = 0$

$$v = -24e^{-2t} + 24$$

$$v = -24e^0 + 24$$

$$= -24 + 24$$

$$\therefore v = 0$$

$\therefore$ initially at rest.

$$\text{ii} \quad v = -24e^{-2t} + 24$$

$$\text{when } t = \ln\left(\frac{4}{3}\right)$$

$$v = -24e^{-2\ln\frac{4}{3}} + 24$$

$$= -24e^{\ln\left(\frac{4}{3}\right)^{-2}} + 24$$

$$= -24e^{\ln\left(\frac{3}{4}\right)^2} + 24$$

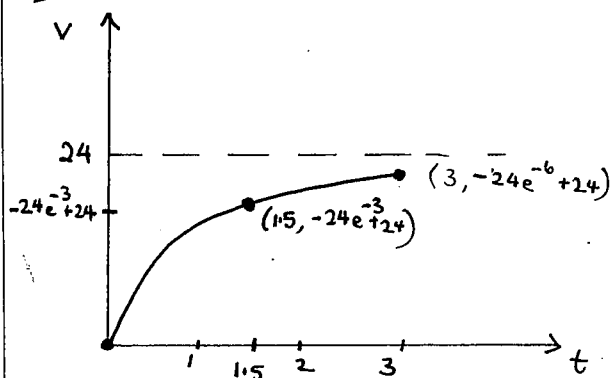
$$= -24e^{\ln\frac{9}{16}} + 24$$

$$= -24\left(\frac{9}{16}\right) + 24$$

$$v = 10\frac{1}{2} \text{ m/s.}$$

$$\text{iii} \quad \text{as } t \rightarrow \infty \quad v \rightarrow 24$$

iv



Academic Year		Calendar Year	
Course		Name of task/exam	

v

t	0	1.5	3
v	0	$-24e^{-3} + 24$	$-24e^{-6} + 24$

distance = area under curve

$$= \int_0^3 f(x) dx$$

$$= \frac{h}{3} [y_0 + y_n + 4y_1]$$

$$= \frac{1.5}{3} [0 + (-24e^{-6} + 24) + 4(-24e^{-3} + 24)]$$

$$= 57.58047...$$

$$= 57.6 \text{ m}$$

vi $a = \frac{dv}{dt}$

$$= 48e^{-2t}$$

we know $v = -24e^{-2t} + 24$

$$\therefore 24e^{-2t} = 24 - v \dots \textcircled{1}$$

$$\therefore a = 2 \times 24e^{-2t}$$

$$a = 2[24 - v] \text{ from } \textcircled{1}$$

$$\therefore a = 48 - 2v$$

Question 16:

$$a \int_{\frac{\pi}{4}}^{\frac{\pi}{2}} \cot x dx = A \log_e B$$

$$\int_{\frac{\pi}{4}}^{\frac{\pi}{2}} \cot x dx = \int_{\frac{\pi}{4}}^{\frac{\pi}{2}} \frac{\cos x}{\sin x} dx$$

$$= [\ln(\sin x)]_{\frac{\pi}{4}}^{\frac{\pi}{2}}$$

$$= \ln\left(\sin \frac{\pi}{2}\right) - \ln\left(\sin \frac{\pi}{4}\right)$$

$$= \ln 1 - \ln \frac{1}{\sqrt{2}}$$

$$= 0 - \ln\left(\frac{1}{\sqrt{2}}\right)$$

$$= \ln\left(\frac{1}{\sqrt{2}}\right)^{-1}$$

$$= \ln \sqrt{2}$$

$$= \ln 2^{\frac{1}{2}}$$

$$= \frac{1}{2} \ln 2.$$

$$\therefore A = \frac{1}{2} \quad B = 2.$$

b $f(x) = \frac{\sec^2 x}{1 + \tan x}$

i y-intercept: set $x = 0$

$$y = \frac{1}{\cos^2(0)[1 + \tan 0]}$$

$$= \frac{1}{1(1)}$$

$$\therefore y = 1$$

$$f(\pi) = \frac{1}{(\cos \pi)^2 [1 + \tan \pi]}$$

$$= \frac{1}{(-1)^2(1)}$$

$$\therefore f(\pi) = \frac{1}{1}$$

Academic Year		Calendar Year	
Course		Name of task/exam	

$$\begin{aligned}
 \text{ii) } \frac{dy}{dx} &= \frac{vu' - uv'}{v^2} & u &= \sec^2 x \\
 & & u' &= 2\sec x (\sec x \tan x) \\
 & & u' &= 2\sec^2 x \tan x \\
 & & v &= 1 + \tan x \\
 & & v' &= \sec^2 x \\
 &= \frac{(1 + \tan x) 2\sec^2 x \tan x - \sec^4 x}{(1 + \tan x)^2} \\
 &= \frac{2\sec^2 x \tan x + 2\sec^2 x \tan^2 x - \sec^4 x}{(1 + \tan x)^2} \\
 &= \frac{\sec^2 x (2\tan x + 2\tan^2 x - \sec^2 x)}{(1 + \tan x)^2} \\
 &= \frac{\sec^2 x (2\tan x + 2\tan^2 x - (1 + \tan^2 x))}{(1 + \tan x)^2} \\
 &= \frac{\sec^2 x (2\tan x + 2\tan^2 x - 1 - \tan^2 x)}{(1 + \tan x)^2} \\
 &= \frac{\sec^2 x (\tan^2 x + 2\tan x - 1)}{(1 + \tan x)^2}
 \end{aligned}$$

iii) For stationary points $\frac{dy}{dx} = 0$

$$\sec^2 x (\tan^2 x + 2\tan x - 1) = 0$$

$$\sec^2 x = 0 \quad \tan x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

no soln

$$= \frac{-2 \pm \sqrt{4 + 4}}{2}$$

$$= \frac{-2 \pm 2\sqrt{2}}{2}$$

$$\tan x = -1 \pm \sqrt{2}$$

$$\therefore x = 0.39, 1.96 \quad \left. \begin{array}{l} y = 0.83 \\ -4.83 \end{array} \right\}$$

$$(0.39, 0.83) \quad (1.96, -4.83)$$

x	0.3	0.39	0.4
$\frac{dy}{dx}$	-0.18	0	0.01

∩ $\therefore \min (0.39, 0.83)$

x	1.9	1.96	2
$\frac{dy}{dx}$	4.4	0	-2.4

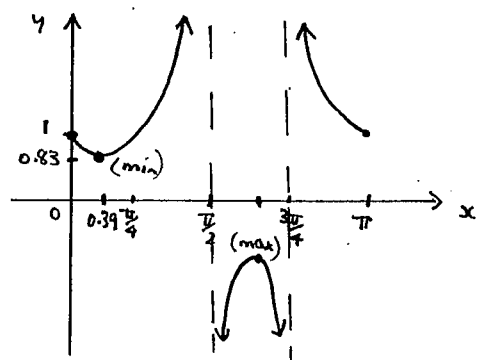
∪ $\therefore \max (1.96, -4.83)$

iv) Vertical asymptotes where den = 0.

$$\begin{aligned}
 \therefore 1 + \tan x &= 0 \\
 \tan x &= -1 \quad 4 \\
 x &= \frac{3\pi}{4}
 \end{aligned}$$

$$\begin{aligned}
 \text{Also } \cos^2 x &= 0 \\
 \cos x &= 0 \\
 x &= \frac{\pi}{2}
 \end{aligned}$$

v



$$\begin{aligned}
 \text{vi) } A &= \int_0^{\pi/3} y \, dx \\
 &= \int_0^{\pi/3} \frac{\sec^2 x}{1 + \tan x} \, dx \\
 &= \left[\ln(1 + \tan x) \right]_0^{\pi/3} \\
 &= \ln(1 + \sqrt{3}) - \ln(1) \\
 &= \ln(1 + \sqrt{3}) \text{ units}^2
 \end{aligned}$$